TABLE OF SPECIFICATIONS

IT 10 — 1st Quarter Examination

S.Y. 2026–2027 · Alan L. Vergara

77-item question bank · 50 items per attempt — **12 required items appear on every attempt** (Part I picture identification + Part IV lab tasks); the remaining 38 are drawn unseen-first from the 65-item optional bank

45 minutes · Multiple Choice & Identification & Rubric-scored Problem Solving · Auto-graded except Part IV (teacher-scored against rubric)

Domain	Content / Learning Competency	Cognitive Skill	Items in Bank	Item Numbers	% of Bank	Per Attempt
A	Arduino Programming Fundamentals — Apply basic Arduino programming constructs (setup/loop, delay, pinMode, digitalWrite, analogRead, serial) to simple control problems	Application / Analysis	25	1–65	32.5%	~14 (of 38)
B	Resistor Color Code — Decode resistor band colours to resistance value and tolerance — every item displays a 4-band diagram	Application	10	16–25	13.0%	~6 (of 38)
C	Resistance Value Computation (Ohm's Law) — Compute resistance from voltage and current using R = V/I	Application / Computation	10	26–35	13.0%	~6 (of 38)
D	Arduino Family & SIK Components — Identification — Identify Arduino board features, pins, and Arduino Starter Kit (SIK) components and their functions	Knowledge / Recall	20	36–55	26.0%	~12 (of 38)
I	Part I — Visual Component Identification — identify SIK/Arduino components from photographs shown in the LMS *(required; items lifted from 1st Quarter Exam.docx)*	Knowledge / Recall (visual)	10	66–75	13.0%	**10 (required)**
IV	Part IV — Problem Solving — write/build in the embedded Virtual Arduino Lab, then submit; graded against the analytic rubric *(required; lab panel opens inside the exam)*	Analysis / Synthesis	2	76–77	2.6%	**2 (required)**
	TOTAL		**77**	1–77	100%	**50**

Test Blueprint — Cognitive Level Distribution

Cognitive Process	Items	% of Bank	Remarks
Knowledge / Recall	30	39.0%	Arduino family & SIK components (20 written + 10 picture-based)
Application / Computation	45	58.4%	Programming constructs, colour code decoding, Ohm's law
Analysis / Synthesis (rubric)	2	2.6%	Part IV problem solving in the Virtual Arduino Lab
Total	**77**	100%	

Generated 2026-10-09 · Source: 1st Quarter Exam.docx + Arduino/Resistor lecture materials · Bank seeded in MIOW LMS (50-item unseen-first draw; Part I + Part IV on every attempt; resistor items show band diagrams)